Personal Document & File Intelligence System — Literature Review Table

Research Title, Author(s)/Publisher, Date, Conclusion, and Research Gap for each paper identified during the literature review. "Not reported" indicates the field could not be verified from the source consulted.

Research Title	Author(s)	Publisher / Venue	Date	Conclusion	Research Gap
The Semantic Desktop – a Basis for Personal Knowledge Management	Sauermann, L.	I-KNOW 2005 (DFKI)	2005	Proposes integrating email, browser and office documents under a shared ontology so files can be searched by meaning rather than filename; introduces the Gnowsis/PIMO concept.	Foundational only — single desktop, no cross-device/cloud, no security dimension.
The Personal Knowledge Workbench of the NEPOMUK Semantic Desktop / SemNotes	NEPOMUK Project consortium	Springer LNCS / ACM Proc. 7th Int'l Conf. on Semantic Systems	2007–2011	Delivers working prototypes for cross-application personal information integration and semantic note-taking using RDF on a single machine.	Confined to one desktop OS; no cross-cloud or mobile coverage; no lifecycle or security features.
The ubiquitous digital file: A review of file management research	Dinneen, J. D., Julien, C. A.	Journal of the Association for Information Science and Technology (JASIST), 71(1), E1–E32	2020	Systematic review of how people actually organize and retrieve personal files; consolidates two decades of PIM/file-management research.	Review identifies persistent gaps in understanding multi-device/multi-location file behavior; does not address security or expiry.
A Systematic Review of Factors Effecting Personal Information Management Using Desktop and Mobile Devices	Ali, I., Warraich, N. F., Butt, K.	Journal of Librarianship and Information Science (Sage), PRISMA-based review	2025/2026	Identifies behavioral and contextual factors affecting PIM across desktop and mobile devices.	Behavioral/usage focus only; no technical architecture for cross-platform indexing or security integration.
NubiVis: A Personal Cloud File Explorer	Not reported (ResearchGate listing)	Not reported	Not reported (~2014 indexed)	Prototype web-based tool for exploring/managing personal files stored in arbitrary locations with a unified interface.	Single system paper, not a sustained research programme; no security, expiry, or malware-awareness layer.
Malware detection in PDF and office documents: a survey	Singh, P., Tapaswi, S., Gupta, S.	Information Security Journal: A Global Perspective, 29(3)	2020	Surveys signature-based, heuristic, behavior-based and ML-based approaches to malicious PDF/Office document detection.	Focuses purely on detection technique taxonomy; no integration with personal file management, location, or lifecycle context.
Malicious PDF detection Based on Machine Learning with Enhanced Feature Set	Yerima, S. Y. et al.	IEEE 14th Int'l Conf. on Computational Intelligence and Communication Networks (CICN)	2022	Proposes an enhanced static-feature ML approach; ensemble SVM ~97.3% and CNN ~99.93% accuracy on CIC Evasive-PDFMal2022.	High accuracy on known evasive samples, but authors note evasive/adversarial PDFs engineered against known features remain the harder case.
Explainable Ensemble Learning Based Detection of Evasive Malicious PDF Documents	Not fully reported (MDPI author list)	Electronics (MDPI), 12(14), 3148	2023	Ensemble learning with enhanced static features, explainable via SHAP; tested against reverse-mimicry injection attacks.	Robustness improves but is not complete against all adversarial injection strategies; single-format (PDF) focus.
UFADF: A Unified Feature Analysis and Detection Framework for Malicious Office Documents	Not reported	IEEE 22nd Int'l Conf. on Trust, Security and Privacy in Computing and Communications (TrustCom)	2023	Extracts comprehensive indicator sets from Office documents and feeds traditional classifiers (RF, SVM, XGBoost, MLP).	Office-document-specific; not integrated with broader personal file management or location intelligence.
uitPDF-MalDe: Malicious PDF files detection using multi machine learning models	Not reported	ScienceDirect (Engineering Applications of Artificial Intelligence)	2025	Compares signature-based, heuristic, and behavior-based detection; proposes a multi-model ML detection pipeline for PDFs.	Notes zero-day/unpatched-vulnerability exploitation remains an open, unresolved risk category for all detection families.
Machine learning-based detection method for malicious PDF files: A temporal classification approach	Not reported	ScienceDirect	2025	Introduces temporal classification for malicious PDFs; cites Check Point 2024 report that 88% of malicious email attachments in 2023 were non-executable files (PDF is #2 abused format after HTML).	Non-executable/document-based attacks are rising faster than defenses adapted for the personal, non-enterprise user segment.
A Survey on Ransomware: Evolution, Taxonomy, and Defense Solutions	Oz, H. et al.	ACM Computing Surveys, 54(11s), Article 238	2022	Comprehensive taxonomy of ransomware defenses; covers file-system-behavior detectors (e.g., ShieldFS) and network-based detectors.	Behavior/entropy-based monitoring detects ransomware only after some file operations (encryption) have already begun — the 'before damage' problem remains only partially solved.
RWGuard: A real-time detection system against cryptographic ransomware	Mehnaz, S., Mudgerikar, A., Bertino, E.	RAID 2018 (Springer LNCS)	2018	Real-time, host-based detection system aiming to minimize the window between ransomware activity and detection.	Reduces but does not eliminate the detection-before-damage gap; host-based, not evaluated across personal multi-cloud storage.
NapierOne: A modern mixed file data set alternative to Govdocs1	Davies, S. R., Macfarlane, R., Buchanan, W. J.	Forensic Science International: Digital Investigation, 40, 301330	2022	Introduces a modern, realistic, mixed-file-type dataset for ransomware/forensics research, replacing the dated Govdocs1 corpus.	Dataset itself is a resource, not a system; highlights the lack of realistic personal-file-mix benchmarks generally.
Ransomware Detection Using Machine Learning: A Survey	Not reported	MDPI (journal: Big Data and Cognitive Computing)	2023	Surveys ML/behavioral ransomware detection studies from 2017–2022; documents established baseline-deviation methodology.	Consistently flags false-positive/false-negative tradeoffs and generalization to new ransomware families as open problems.
Data Loss Prevention (DLP) for Documents — vendor/technical overview	Palo Alto Networks / Check Point (technical documentation, not peer-reviewed)	Vendor documentation	2024–2026 (ongoing)	Describes regex/EDM/IDM content-matching, OCR+ML classification, and policy-based blocking used in enterprise DLP systems.	Entire DLP paradigm assumes a central IT policy server and admin oversight — no adaptation exists for a single consumer device with no such infrastructure.
Fingerprinting of Machines in Critical Systems for Integrity Monitoring and Verification	Not fully reported (arXiv preprint)	arXiv:2412.16595	2024	Reviews file-integrity-monitoring lineage from Tripwire (Kim & Spafford, 1993) onward; cites Peddoju et al. (2020) on unresolved FIM problems (delayed detection, deployment complexity, tamper resistance).	FIM techniques remain enterprise/server-oriented; no fixed 'trusted baseline' concept exists for actively-edited personal documents.
Ensuring Data Integrity in Storage: Techniques and Applications	Stony Brook University, File Systems and Storage Lab (FSL)	FSL technical survey	Not reported	Classifies integrity-assurance techniques by scope, logical layer, and checking mode (mirroring, parity, checksumming, cryptographic hashing).	Purely technique-level; no application to personal document lifecycle, location, or a composite personal risk metric.
Cross-Platform Analysis of Indirect File Leaks in Android and iOS Applications	Not fully reported	arXiv:1511.00104	2015	Shows that despite sandbox-based app isolation on Android/iOS, indirect file leaks between apps remain a serious, unsolved threat.	Confirms mobile sandboxing is a structural barrier to any app (malicious or legitimate) achieving broad cross-app file visibility.
Unauthorized Cross-App Resource Access on MAC OS X and iOS	Not fully reported	arXiv:1505.06836	2015	Details TrustedBSD MAC-based sandboxing on macOS/iOS; each app confined to its own container unless explicit entitlements are granted.	Establishes the hard OS-level ceiling on any consumer app's ability to silently index a user's full device across Apple platforms.

Research Title	Author(s)	Publisher / Venue	Date	Conclusion	Research Gap
PolyScope: Multi-Policy Access Control Analysis to Triage Android Scoped Storage	Not fully reported	arXiv:2302.13506	2023	Analyzes how Android Scoped Storage (Android 10+) restricts app access to package-private and shared storage directories.	Confirms scoped storage is a genuine, evolving constraint — full-device file location tracking on Android needs the restricted MANAGE_EXTERNAL_STORAGE permission, generally reserved for file-manager/backup apps.
A Comprehensive Review on Cybersecurity of Digital Twins Issues, Challenges, and Future Research Directions	Not fully reported	IEEE Xplore	2024	Systematic review of digital-twin cybersecurity across component/process/asset/system/network-of-systems levels.	Entirely industrial/CPS/IoT-focused — no literature ties 'digital twin' to an individual person's personal document set; a genuine conceptual gap if used carefully.

Notes on sourcing

This table condenses the papers identified in the accompanying deep literature review. Several arXiv preprints and 2023–2026 journal articles did not expose full author lists in the search results consulted; these are marked "Not reported" rather than guessed. DOIs and full citation details should be independently verified against the publisher's page before use in a formal submission. Vendor/technical documentation (e.g., DLP overviews) is included for completeness but is not peer-reviewed academic literature and should be labelled as such in any bibliography.